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Quantum lattice Boltzmann method for simulating nonlinear fluid dynamics

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Quantum computing holds transformative potential for simulating nonlinear physical systems, such as fluid turbulence. However, mapping nonlinear dynamics to the linear operations required by quantum hardware remains a fundamental challenge. Here we bridge this gap by introducing a novel node-level ensemble description of a lattice gas, which enables the simulation of nonlinear fluid dynamics on quantum computers. This approach combines the advantages of the lattice Boltzmann method with low-dimensional representation (computational cost) and lattice gas cellular automata with linear collision treatment (quantum compatibility). Building on this framework, we propose a quantum lattice Boltzmann method that relies on linear operations with medium dimensionality, offering the potential for quantum speedup. We validated the algorithm through simulations of vortex-pair merging and decaying turbulence on up to 16.8 million computational grid points. The results demonstrate remarkable agreement with direct numerical simulation, effectively capturing the essential nonlinear mechanisms of fluid dynamics. This work potentially advances the development of quantum algorithms for other nonlinear problems across various transport phenomena in engineering.

INTRODUCTION

Classical numerical methods for simulating fluid dynamics and general nonlinear problems demand substantial computational resources. For instance, the direct numerical simulation (DNS) of turbulence faces significant challenges due to the cubic growth of computational cost with the Reynolds number (Re), which is necessary for resolving multiscale flow structures [1, 2]. Quantum computation, leveraging the principles of superposition and entanglement inherent in quantum states, has shown an exponential advantage over classical computational methods in certain specific problems [3–5]. Consequently, quantum computing for fluid dynamics (QCFD) [6–12] is anticipated to tackle turbulence, one of most challenging problems in classical physics [13] and nonlinear dynamics.

Classical and quantum systems exhibit intrinsic differences: fluid dynamics is nonlinear and irreversible, whereas quantum computing relies on linear and unitary operations. This fundamental disparity presents a substantial challenge in simulating nonlinear dynamics using quantum computers [14, 15]. Several approaches have been proposed to partially address this challenge. The Navier-Stokes (NS) equations governing fluid motion have been transformed into the Schrödinger equation and solved via Hamiltonian simulation [8, 10, 16]. They have also been discretized into a set of nonlinear ordinary differential equations and then linearized [7, 17]. Variational quantum algorithms [18] offer a feasible approach for addressing the nonlinearity in fluid dynamics simulations [19, 20]. Quantum-classical hybrid methods have been employed to solve these equations [6, 21–24]. Moreover, the general partial differential equations can be transformed into linear ones by increasing the dimensionality of the problem [25–28].

In particular, the earliest attempt [29] in QCFD used the lattice Boltzmann method (LBM) [30, 31], which provides a natural framework for addressing nonlinearity in fluid dynamics. By reformulating the NS equations into a higher-dimensional

lattice gas representation, LBM inherently transforms the nonlinear convection term in physical space into a linear transport term in phase space. However, the collision term in LBM, usually treated by the Bhatnagar-Gross-Krook (BGK) model [32], introduces additional nonlinearity, posing a significant challenge for its quantum implementation.

Therefore, developing a linear yet physically accurate collision model is crucial for the quantum lattice Boltzmann method (QLBM) [9, 11, 33]. In existing studies, the nonlinear part of the collision term has often been neglected to simulate simple convection-diffusion problems [34–41]. The nonlinearity in the collision term has been eliminated using the Carleman linearization, but this approach results in an exponential increase in the dimension of the linearized system with scale [11, 42–44]. Another collision model in the lattice-gas cellular automata (LGCA) [45–47] is linear and can be integrated into QLBM [48, 49], but its quantum algorithms suffer from excessively high dimensionality [50]. Consequently, an efficient QLBM algorithm for simulating nonlinear fluid dynamics has yet to be developed.

Here we introduce a QLBM algorithm with a novel ensemble description of lattice gas to solve the NS equations. This ensemble assumes that each computational node contains multiple realizations, with all nodes being mutually independent. Our QLBM combines the strengths of both the LBM and LGCA, featuring medium dimensionality and linear collision treatment [46]. In the implementation, the probability distribution of the ensemble is encoded into amplitudes of quantum states. The evolution of the quantum states is driven by linear transport and collision steps. Additionally, an H-step is employed to keep the ensemble near thermodynamic equilibrium by reducing artificial correlations between qubits (measured by quantum entropy). The flow field is measured only at the end of simulation, thereby avoiding the need for frequent intermediate measurements and state preparations.

Notably, our QLBM is validated through simulations of vortex-pair merging and turbulence on up to 16.8 million computational grid points, representing the largest-scale case in QCFD to date. For comparison, the size of most existing QCFD simulations is limited to roughly 1,000 grid points. This

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successful application demonstrates the capability of QCFD to address highly nonlinear flow prediction, and provide insights into developing quantum algorithms for other nonlinear problems in various engineering applications.

RESULTS

Method overview

Fluid dynamics is nonlinear, while quantum computing requires linearity. The key idea of our proposed QLBM is to ‘lift’ a 2D/3D fluid system into a higher-dimensional (but still efficient) linear lattice gas model. This allows quantum hardware to handle nonlinearity via ensemble transformations, while an entropy-stabilizing step ensures physical equivalence between the quantum state and classical fluid dynamics.

The QLBM is used to solve the Boltzmann equation, which describes the evolution of the statistical distribution function of a particle system, and reduces to the NS equations at a small Knudsen number. Therefore, our QLBM holds the potential to simulate nonlinear problems across a wide range of transport phenomena in engineering.

In classical LBM or LGCA, the Boltzmann equation is solved using lattice gas models by discretizing the particle velocity into q discrete values through various DdQq models [31] within a node in a d -dimensional space. As illustrated in Fig. 1, the lattice $\mathcal{L}$ comprises N discrete nodes (black dots). Each node represents a local flow state at a spatial location $\mathbf{x} = (x, y)$ and stores particle properties. The particles transport from one node to another along q predefined discrete directions (arrows), and they collide at the nodes. The collision of particles within a node drives the local lattice gas toward an equilibrium state [47]. Each node consists of q cells (red circles), and each cell can be either empty (open circles) or occupied (closed circles) by at most one particle with a specified velocity.

As illustrated in Figs. 1(a) and (c), neither classical LBM nor LGCA can simultaneously achieve manageable dimensionality and involve linear particle collisions. To address this, we propose a meso-level ensemble, situated between the LBM and LGCA, as shown in Fig. 1(b). This ensemble forms the basis of our QLBM, resulting in medium dimensionality and linear collisions within the quantum algorithm.

The corresponding QLBM algorithm is sketched in Fig. 2. Initially, the flow field is encoded into the quantum state $|\Psi\rangle$. Then the quantum state evolves through transport and collision steps at the node level. When the state begins to deviate from equilibrium, an H-step is invoked to relax the state back to equilibrium. Upon reaching a specified time, the quantum state is measured to reconstruct the entire flow field.

Ensemble description for QLBM

The present QLBM employs the probability encoding method to embed the flow field into the amplitudes of quantum states within the lattice gas framework. As illustrated in Fig. 1, we use the concept of the Gibbs ensemble [51] to re-describe LBM and LGCA from the perspective of statistical mechanics at the velocity, node, and lattice levels, which facilitates developing

an efficient quantum algorithm for our proposed lattice gas system.

The velocity-level ensemble, named after the velocity distribution, corresponds to classical LBM. In this ensemble, each realization within a node is illustrated by a ‘‘card’’ (containing a node) in Fig. 1(a). Each node is independent, and it involves q independent realizations or cells [50] with different velocity directions. Consequently, the dimensionality of this ensemble is $N_d = Nq$. Each realization at $\mathbf{x}$ and time t occurs with the probability $p_j^{(n_j)}(\mathbf{x}, t)$, where the subscript $j = 1, \dots, q$ denotes for a quantity along the j -th direction, and the Boolean variable $n_j(\mathbf{x}, t)$ describes the occupation status of cells: $n_j = 1$ indicates the presence of a particle with velocity $\mathbf{e}_j$, and $n_j = 0$ indicates the absence of such a particle. The ensemble average of n_j yields the velocity distribution function [50]

$$f_j(\mathbf{x}, t) = \sum_{n_j=0}^1 n_j p_j^{(n_j)} = p_j^{(1)}. \quad (1)$$

On each card, the cells are color-coded by $p_j^{(1)}(\mathbf{x}, t)$. Moreover, the fluid density ρ and velocity $\mathbf{u}$ are defined as $\rho = \sum_{j=1}^q f_j$ and $\rho \mathbf{u} = \sum_{j=1}^q f_j \mathbf{e}_j$.

The lattice-level ensemble, corresponding to LGCA, is a collection of all $N_d = 2^q N$ possible states of the entire lattice, and all N nodes in the lattice are fully correlated. Note that the excessively high N_d in LGCA causes its quantum algorithm to lose speedup advantage. In Fig. 1(c), each realization for cellular automata is illustrated by a card with multiple nodes, and closed and open circles denote occupied and empty cells, respectively. At the lattice level, a realization is represented as $\bigotimes_{\mathcal{H}} \mathbf{n}(\mathbf{x})$, where $\mathbf{n} = \{n_1, \dots, n_q\}$ denotes the state for each node, and $\bigotimes$ the Kronecker product of vectors over the Hilbert space $\mathcal{H}$ of node coordinates. Each card is color-coded by the probability $p_{\bigotimes_{\mathcal{H}} \mathbf{n}(\mathbf{x})}$ of each realization.

The node-level ensemble, between the velocity- and lattice-level ones, combines the low-dimensional representation of LBM with the linear collision treatment of LGCA. In this ensemble, each node accommodates 2^q realizations of $\mathbf{n}$, and nodes are independent of each other, unlike the lattice-level one where nodes are fully correlated. This key property leads to medium $N_d = 2^q N$. At the node level, we define the probability $p_{\mathbf{n}}(\mathbf{x})$ of $\mathbf{x}$, satisfying $\sum_{\mathbf{n}} p_{\mathbf{n}} = 1$ within a node, and the joint probability $p_{\mathbf{x}, \mathbf{n}}$ of $\mathbf{x}$ and $\mathbf{n}$, satisfying $\sum_{\mathbf{x}, \mathbf{n}} p_{\mathbf{x}, \mathbf{n}} = 1$ over a lattice. Each card for realization $\{\mathbf{x}, \mathbf{n}\}$ is color-coded by $p_{\mathbf{x}, \mathbf{n}}$ in Fig. 1(b). The independence of nodes leads to $p_{\mathbf{x}, \mathbf{n}} = p_{\mathbf{n}} \rho / \sum_{\mathbf{x}} \rho$.

Algorithm of QLBM

The QLBM integrates certain components from both classical LBM and LGCA (see Methods). As illustrated in Fig. 2, all realizations $\{\mathbf{x}, \mathbf{n}\}$ at the node level (blue panels) are encoded into a quantum state $|\Psi\rangle$, and $|\Psi\rangle$ is evolved in QLBM with ensemble transformation using a prediction-correction approach.

The prediction phase involves transport and collision steps, which are reformulated from those in LGCA to the node level.

The transport step

$$\{\mathbf{x}, \mathbf{n}\} \mapsto \sum_{j=1}^q \frac{n_j}{\rho_n} \{\mathbf{x} + \mathbf{e}_j, \mathbf{n}\} \quad (2)$$

is adapted from Eq. (9), where $\mapsto$ denotes a mapping and $\rho_n(\mathbf{x}, \mathbf{n}) \equiv \sum_{j=1}^q n_j$ denotes the number of particles at a node. In Eq. (2), the realization $\{\mathbf{x}, \mathbf{n}\}$ is divided into ρ_n components, and each component uniformly propagates into ρ_n neighboring nodes with the equal proportion n_j/ρ_n .

In the collision step, each possible collision event is encoded among the realizations in the ensemble, as illustrated by a collection of cards for a node in Fig. 1(b). This allows the LGCA collision model [46, 47], originally developed at the lattice level, to be effectively used at the node level. Incorporating the collision step in Eq. (10) into the node-level realization $\{\mathbf{x}, \mathbf{n}\}$ yields the mapping $\{\mathbf{x}, \mathbf{n}\} \mapsto \{\mathbf{x}, \mathbf{n}'\}$ from the initial state $\mathbf{n}$ to the post-collision state $\mathbf{n}' = \{n'_1, \dots, n'_q\}$. Incorporating the collision probability $\gamma \in [0, 1]$ into this mapping leads to a linear operation

$$\{\mathbf{x}, \mathbf{n}\} \mapsto \gamma \{\mathbf{x}, \mathbf{n}'\} + (1 - \gamma) \{\mathbf{x}, \mathbf{n}\}, \quad (3)$$

where γ governs the fluid viscosity. Note that the transport and collision steps are linear, positive, L^1 -norm preserving, and numerically stable (see Secs. 2 and 4 in Supplementary Information), and their complexities are discussed in Methods. The linear operations can be efficiently represented as unitary transformations in quantum algorithms, and the positivity and L^1 -norm preserving guarantee that the probability distribution remains physically realizable.

During these linear transport and collision steps, the cells within a node can become correlated. This contradicts to the cell independence constraint at the velocity level, and causes the deviation of f_j from the equilibrium velocity distribution $f_{eq,j}$. This deviation, quantified by the H-function (an entropy measure), results in unphysical results of QLBM (see Sec. 3 in Supplementary Information). Thus, the correction phase in the QLBM algorithm introduces the H-step to keep f_j near equilibrium, through inter-ensemble transformations between velocity and node levels across different N_d .

The node-to-velocity-level transformation is through a linear and surjective mapping

$$p_j^{(n_j)} = \sum_{m=1}^q \sum_{n_m=0}^1 p_n \quad (4)$$

based on Eq. (1) and mass conservation. However, the inverse transformation is non-unique, because $N_d = N2^q$ at the node level is much higher than $N_d = Nq$ at the velocity level. Consistent with the independent cells within a node, we establish the velocity-to-node-level transformation

$$p_n = p_{\text{ind},n} \equiv \prod_{j=1}^q p_j^{(n_j)}, \quad (5)$$

where the subscript “ind” denotes the independence of indi-

vidual $p_j^{(n_j)}$ components. This choice ensures that all possible p_n satisfy Eq. (4). It is also consistent with the form of the local equilibrium probability [52] at a node, facilitating relaxing the ensemble back to equilibrium.

As illustrated in Fig. 3, the H-step applies Eqs. (4) and (5), relaxing distribution of p_n toward the local equilibrium one of $p_{\text{ind},n}$, along with the growth of quantum entropy (see Sec. 3 in Supplementary Information). In the quantum representation, each cell at the node level is encoded by a single qubit. Entangled states correspond to cells that are not independent of each other [53], while separable states correspond to independent cells. The H-step is equivalent to disentangling the qubits without changing the measurement outcomes. This challenging quantum separability problem can hinder the overall speedup of the QLBM algorithm (see Eq. (16) and relevant discussion in Methods), and is expected to be resolved [54, 55] in the future work.

In the implementation of the QLBM algorithm, the probabilities are encoded into quantum states, and the states are evolved using quantum circuits (see Methods and Sec. 2 in Supplementary Information). The sensitivity of the quantum circuit to noise is analyzed in Sec. 7 in Supplementary Information.

Validation of QLBM

We assess the proposed QLBM algorithm [50] using three typical 2D incompressible flows, the Taylor-Green (TG) vortex, vortex-pair merging, and decaying turbulence. In post-processing of simulation data, fluid density and velocity are calculated from f_j , and f_j is calculated from Eqs. (1) and (4) where the probability is measured from the quantum state (see Sec. 2E in Supplementary Information).

The QLBM simulations were conducted with the D2Q9 model (see Methods). Note that the NS equation

$$\frac{\partial}{\partial t}(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla P + \nabla \cdot (\mu \nabla \mathbf{u}) \quad (6)$$

can be derived from the D2Q9 model with the LGCA collision step in Eq. (3) [56], where $P = \sum_{j=1}^q (\mathbf{e}_j - \mathbf{u}) \cdot (\mathbf{e}_j - \mathbf{u}) f_j / 3$ denotes the pressure and μ the viscosity. It is straightforward to extend the QLBM to simulating 3D flows with more sophisticated DdQq models and more computational resources.

The QLBM of the TG vortex was conducted at $\text{Re} = 13$ and 27 on 128^2 and 256^2 nodes, respectively, where the viscosity is calculated from Eq. (15) with $\gamma = 0.5$. This benchmark problem has the initial velocity $\mathbf{u}_0 = (\sin x \cos y, -\cos x \sin y) / 8$ and the exact solution of vorticity $\omega = e^{-2\nu t} \sin x \sin y / 4$, showing the exponential decay of vortex intensity. The excellent agreement of the QLBM results and the exact solutions at different Re in Fig. S1 indicates that the QLBM accurately captures viscous dissipation.

QLBM simulation of vortex-pair merging

Merging of a pair of vortices involves nonlinear vortex interactions, which are crucial for many practical applications in vortical flows and turbulence. The initial vorticity is the superposition of the two Burgers vortices separated by a distance l . Each vortex is characterized by a Gaussian vorticity distribu-

tion $\omega = (\Gamma/2\pi\sigma^2) \exp(-r_v^2/2\sigma^2)$, with the radial distance r_v from the vortex center. The QLBM of the vortex-pair merging was conducted at $\text{Re} = 350$ on 2048^2 nodes, with $\sigma = 0.1\pi$, $\Gamma = 0.2\pi$, and $l = 0.59$. Figure 4(a) illustrates the mutual rotation of two vortices driven by their induced velocities. The vortices first undergo persistent stretching and deformation, and then nonlinear vortex interaction and viscous diffusion initiate the merging process. Finally, the two vortices are merged into one with two spiral-like arms.

The QLBM result is assessed using the result of the direct numerical simulation (DNS) for incompressible flow with the pseudo-spectral method [57, 58]. Figure 4(b) demonstrates quantitative agreement between the vorticity distributions from QLBM and DNS, along the $y = \pi$ cross-section of the 2D vorticity field, where the migration of peaks in these distributions characterizes the vortex-pair merging process. The deviation of the velocity $\mathbf{u}$ in QLBM simulation from that $\mathbf{u}_{\text{DNS}}$ in DNS is quantified by the normalized L^2 -norm error $\mathcal{R} \equiv \|\mathbf{u} - \mathbf{u}_{\text{DNS}}\|_2 / \|\mathbf{u}_{\text{DNS}}\|_2$. In general, our QLBM can accurately simulate nonlinear dynamics of vortex interactions.

Furthermore, the discrete vorticity from QLBM is transformed into a continuous probability distribution using the kernel density estimation (KDE). The correlation coefficient r of normalized vorticities from QLBM and DNS decreases with time in Fig. 4(c). The KDE analysis reveals that slight discrepancies between QLBM and DNS in the low vorticity region. This discrepancy may stem from the deviation from equilibrium in the lattice gas system [32], which is attributed to the lack of growth of the H-function during the collision step. Here we selected $\gamma = 1$ to achieve the largest Re (see Eq. (15)) to clearly show the nonlinear vortex interaction, but this choice also causes a milder growth of the H-function during the collision step (see Sec. 3 in Supplementary Information).

QLBM simulation of decaying turbulence

The quantum computing of turbulence is considered to be extremely challenging [8, 15, 59], because turbulence is highly nonlinear whereas quantum computing is based on linear operations. The present QLBM of turbulence showcases the capability of quantum computing for simulating one of the most challenging problems in classical physics [13].

The velocity of 2D decaying homogeneous isotropic turbulence is initialized using a Gaussian random field on 4096^2 nodes with the spatially averaged initial velocity magnitude $|\mathbf{u}_0| = 0.03$ and a specified energy spectrum [60], $E_k(k) \sim k^{-5/3} f_L(k)$ at $k < \sqrt{N}/3$, where k is the wavenumber in Fourier space, and $f_L = (k/(k^2 + 4))^{1/2} k^{5/3+2}$ characterizes the shape of E_k in the energy-containing range. The maximum velocity magnitude $0.12 \ll 1$ satisfies the low-Mach-number condition [31]. Here $\text{Re} = 102$ is small, due to the small $|\mathbf{u}_0|$ and the medium viscosity, which is proportional to $1/\sqrt{N}$ with modest N and D2Q9 model [61]. To increase Re , a larger N and a more sophisticated DdQq model can be employed.

The vorticity evolution in Fig. 5(a) exhibits consistent decaying rates and morphological characteristics between DNS and QLBM results, including the dissipation of small-scale flow structures and the generation of large-scale ones. In Fig. 5(b), the decay of the average kinetic energy $|\mathbf{u}|^2/2$ in QLBM is

slightly slower than that in DNS, perhaps due to the insufficient dissipation mechanisms at high wavenumbers in QLBM. The multiscale features and energy cascade are characterized using the energy spectrum in Fig. 5(c). Consistent with the vorticity contour in Fig. 5(a), the low-wavenumber region of E_k remains nearly constant over time, while the high-wavenumber region dissipates gradually, representing the viscous dissipation and inverse energy cascade at small scales [62]. The slightly larger E_k in QLBM than DNS at high wavenumbers indicates that QLBM has a slower energy transfer rate than DNS.

DISCUSSION

We propose the QLBM algorithm with ensemble transformations for simulating nonlinear fluid dynamics problems, including turbulence. The significant challenge in quantum computing for solving such highly nonlinear problems is addressed by two conceptual steps. First, we transform a low-dimensional, nonlinear fluid system to a medium-dimensional, linear lattice gas system. The resultant node-level ensemble permits linear operations for particle transport and collision in a quantum algorithm. Second, the transformation between ensembles at different dimensions can cause inconsistent physical constraints, leading to the deviation of the velocity distribution from its equilibrium state. Thus we introduce the H-step to maintain the velocity distribution near equilibrium, which is equivalent to disentangling qubits without altering the measurement outcomes of each qubit.

Our QLBM algorithm is linear, positive, and L_1 -norm preserving. It can be implemented on a quantum computer without the need for classical computation, frequent intermediate measurements, or repetitive state preparation processes. Furthermore, it shows potential for speedup in large-scale flow simulations (see Sec. 6 in Supplementary Information).

The QLBM has been validated by three typical 2D incompressible flows. The QLBM results show an excellent agreement with the exact solution for the linear TG vortex, and good agreements with the DNS results for the highly nonlinear vortex-pair merging and decaying turbulence on up to 4096^2 nodes. These successful applications of the QLBM showcase the capability of quantum computing to simulate highly nonlinear dynamics.

The QLBM incorporating ensemble transformation provides insights into quantum computing for nonlinear problems broadly and QCFD specifically. Integrating the understandings from both nonlinear classical systems and their quantum counterparts is essential for the development of an efficient quantum algorithm.

On the other hand, there are some issues to be addressed in the future work. First, the execution of the H-step demands significant qubit resources and may limit the potential speedup, and thus enhancing the efficiency of the H-step is essential. Second, the QLBM algorithm needs to be further improved for running on quantum devices, e.g., the current QLBM case for 2D turbulence requires $n_Q = 24 + 9 = 33$ qubits to encode the flow field and the quantum circuit depth can reach n_Q^2 for each time step, which poses a significant challenge for the current hardware [63]. Potential challenges on efficient

state preparation/measurement and effects of noises are discussed in Sec. 7 in Supplementary Information. Furthermore, error correction/mitigation methods should be applied in the future work. Third, the QLBM is restricted to low Mach numbers, and has not incorporated no-slip boundary conditions. These issues will be addressed to enhance its applicability to more complex problems, such as multiphase and porous media flows, as well as more general nonlinear dynamics problems beyond fluid dynamics.

METHODS

Classical LBM and LGCA

The Boltzmann equation can be discretized into the lattice Boltzmann equation $f_j(\mathbf{x} + \mathbf{e}_j \delta t, t + \delta t) - f_j(\mathbf{x}, t) = \Omega_j$ at the velocity level and the LGCA equation $n_j(\mathbf{x} + \mathbf{e}_j \delta t, t + \delta t) - n_j(\mathbf{x}, t) = \Omega_j$ at the lattice level, with the time step δt . The collision term Ω_j ensures the conservation of particles and momentum.

Each time step in LBM can be decomposed into the transport step

$$f_j(\mathbf{x}, t) \mapsto f_j(\mathbf{x} + \mathbf{e}_j \delta t, t) \quad (7)$$

and the collision step

$$\begin{aligned} f_j(\mathbf{x} + \mathbf{e}_j \delta t, t) &\mapsto f_j(\mathbf{x} + \mathbf{e}_j \delta t, t) + \Omega_j(\mathbf{x} + \mathbf{e}_j \delta t, f_j) \\ &= f_j(\mathbf{x} + \mathbf{e}_j \delta t, t + \delta t). \end{aligned} \quad (8)$$

Each time step in LGCA can also be divided into the transport step

$$n_j(\mathbf{x}) \mapsto n_j(\mathbf{x} + \mathbf{e}_j \delta t) \quad (9)$$

and the collision step

$$\begin{aligned} n_j(\mathbf{x} + \mathbf{e}_j \delta t, t) &\mapsto n_j(\mathbf{x} + \mathbf{e}_j \delta t, t) + \Omega_j(\mathbf{x} + \mathbf{e}_j \delta t, n_j) \\ &= n_j(\mathbf{x} + \mathbf{e}_j \delta t, t + \delta t). \end{aligned} \quad (10)$$

The collision in LBM is widely treated by the BGK model $\Omega_j = -\kappa(f_j - f_{eq,j})$ with the relaxation time coefficient κ [32]. Particularly, the local equilibrium velocity distribution $f_{eq,j}$ is a nonlinear function of $\mathbf{u}$ and $\mathbf{e}_j$, prompting our attempt to adopt the linear collision model in LGCA. Under the collision rules in Fig. S2 in Supplementary Information, the collision event occurs with probability γ . We characterize the collision using the collision term in LGCA, which can be expressed as a linear combination of products of n_j in the lattice-level ensemble [46]. This property makes it well-suited for implementation in quantum algorithms.

Probability encoding for QLBM

We encode the ensemble description of lattice gas at the node level into a quantum state $|\Psi\rangle$ from a single qubit, through q qubits, to $q + \log_2 N$ qubits, corresponding to encoding probabilities $p_j^{(n_j)}$, $p_{\mathbf{n}}(\mathbf{x})$, and $p_{\mathbf{x},\mathbf{n}}$, respectively. First, we

encode $p_j^{(n_j)}$ at the velocity level using a single-qubit state

$$|f_j\rangle \equiv \sqrt{p_j^{(0)}}|0\rangle + \sqrt{p_j^{(1)}}|1\rangle \quad (11)$$

in a superposition of quantum basis states $|0\rangle$ and $|1\rangle$, where $|0\rangle$ and $|1\rangle$ encode $n_j = 0$ and 1, respectively, and $p_j^{(0)}$ and $p_j^{(1)}$ are probabilities for measuring the qubit in states $|0\rangle$ and $|1\rangle$, respectively.

Second, we encode the node-level probability $p_{\mathbf{n}}$ in Eq. (5) using q qubits for each node. To encode all possible (up to 2^q) states of a node, we employ a q -qubit pure state

$$|f_q \cdots f_1\rangle \equiv \sum_{n_1, \dots, n_q=0}^1 \sqrt{p_{\mathbf{n}}} |n_q \cdots n_1\rangle, \quad (12)$$

where $|n_q \cdots n_1\rangle = \bigotimes_{j=1}^q |n_j\rangle$ denotes basis states, and n_1 and n_q are the lowest and highest indices of the qubit sequence, respectively. As illustrated in Fig. S3(a) in Supplementary Information, we establish an encoding bijection $\mathbf{n} \leftrightarrow |n_q \cdots n_1\rangle$ between the node-level $\mathbf{n}$ and basis states $|n_q \cdots n_1\rangle$. The measurement of $|n_q \cdots n_1\rangle$ yields $p_{\mathbf{n}}$ in Eq. (12), transforming the quantum state into the classical probability. Note that Eq. (12) can be decomposed into a product of q separable qubits

$$|f_q \cdots f_1\rangle = \bigotimes_{j=1}^q |f_j\rangle \quad (13)$$

at the velocity level, when q cells for a node are independent. Consistent with Eq. (5), this encoding method can be transformed between the velocity and node levels when each $|f_j\rangle$ is independent.

Third, we encode the joint probability $p_{\mathbf{x},\mathbf{n}}$ for lattice $\mathcal{L}$. Besides encoding the node state using q qubits, all coordinates $\mathbf{x} \in \mathcal{L}$ are encoded into the basis states by the binary representation $\mathbf{x} \leftrightarrow |\mathbf{x}\rangle$ using $\log_2 N$ qubits. As illustrated in Fig. S3(b) in Supplementary Information, each realization is encoded as $\{\mathbf{x}, \mathbf{n}\} \leftrightarrow |n_q \cdots n_1\rangle |\mathbf{x}\rangle$. Finally, we encode all realizations $\{\mathbf{x}, \mathbf{n}\}$ at the node level into a $(q + \log_2 N)$ -qubit quantum state

$$|\Psi\rangle \equiv \sum_{\mathbf{n}} \sum_{\mathbf{x}} \sqrt{p_{\mathbf{x},\mathbf{n}}} |n_q \cdots n_1\rangle |\mathbf{x}\rangle. \quad (14)$$

Quantum implementation of QLBM

The QLBM algorithm is implemented using quantum circuits. As sketched in Fig. 2, the flow field is firstly transformed into the equilibrium velocity distribution $f_{eq,j}$ based on macroscopic state, and then $f_{eq,j}$ is encoded into the quantum state $|\Psi\rangle$ at the node level using Eqs. (1), (5), and (14). Then, $|\Psi\rangle$ sequentially undergoes the transport step, collision step, and H-step.

In the quantum state preparation, the velocity and density fields are sequentially transformed into $f_{eq,j}$, $p_{\mathbf{x},\mathbf{n}}$, and $|\Psi\rangle$. During the transport step, qubits encoding coordinates undergo phase shifts in different directions, controlled by $|f_j\rangle$. Each

basis is split into ρ_n parts according to Eq. (2). In the collision step, all $|f_j\rangle$ exchange states according to predefined collision rules. Collision occurs at a given probability γ , controlled by an ancilla qubit. In the H-step, f_j is obtained by tracing off all qubits except the one encoding $|f_j\rangle$. The circuit is then repeated q times to obtain all f_j , and all f_j are combined by exchanging cross terms of different coordinates (see Sec. 5 in Supplementary Information). Finally, qubits are measured one by one to obtain f_j . The quantum circuit and implementation detail for each step are provided in Sec. 2 in Supplementary Information.

Note that the complexity of the H-step (see Sec. 6 in Supplementary Information) could be a weakness of this algorithm. It causes linear growth in the number of qubits required for every H-step, and exponential growth in the initialization, transport, and collision steps. A possible improvement to the H-step lies in developing a more efficient and effective disentanglement algorithm (see Sec. 3 in Supplementary Information) that does not require repeating the circuit to prepare all f_j .

Simulation setup of QLBM

All QLBM simulations were performed in a periodic square domain with side 2π , $N = N_x^2$ uniformly distributed nodes, and periodic boundary conditions. The time step is $\delta t = 2\pi/N_x$. The kinematic viscosity $\nu = \mu/\rho$ with $\rho = 1$ is determined by the viscosity

$$\mu = \frac{e}{\gamma_{2D}e^2 + \gamma_{2B}(1-e)^2} \delta t \quad (15)$$

with the internal energy e and the subscript of γ denoting the particle collision type (see Fig. S2 in Supplementary Information). We set $\gamma_{2D} = \gamma_{2B} = \gamma$ and $e = 1/3$ [56] in all simulations. The Reynolds number is defined by $\text{Re} = 2\pi|\bar{u}_0|/\nu$, where $\bar{(\cdot)}$ denotes the spatial average.

The QLBM was implemented using our in-house code [64], which employs the quantum algorithm on a classical desktop computer. Only the diagonal elements of the density matrix for $|\Psi\rangle$ were allocated, requiring the memory of $O(2^{n_Q})$ bytes to realize the QLBM with up to $N_x N_y = 4096^2$ nodes. Note that although the initial and final states can be feasibly accessed on a classical computer, corresponding state preparation and measurement can be costly on a quantum computer, which is discussed in Sec. 7 in Supplementary Information.

Complexity analysis

The computational complexity of the QLBM is analyzed in terms of the number of two-qubit gates (detailed in Sec. 6 in Supplementary Information). The algorithm requires $n_Q = \log_2 N + q$ qubits to encode the flow field. The initialization cost depends on the preparation of the N_M dominant modes, yielding a gate complexity of $O(\log^c(qN_M))$ with $c < 4$. The complexities of each transport step, collision step, and H-step are $O(q^2 \log_2^2 N)$, $O(q)$, and $O(q \log_2 N)$, respectively. The total QLBM cost with N_t time steps and N_H H-steps is $C_{qc} \sim q^{N_H} q^2 \log_2^2 N$, with $N_H \leq N_t$. Compared to the

classical LBM cost $C_{\text{classic}} = NqN_t$, the ratio is

$$\frac{C_{qc}}{C_{\text{classic}}} \sim \frac{q^{1+N_H} \log_2^2 N}{NN_t}. \quad (16)$$

In principle, quantum speedup for the present QLBM can only be achieved under a very strict condition $N \gg q^{N_H}$. Since the H-step dominates the cost, improving its efficiency is essential for practical speedup on a quantum computer. Compared to other QCFD methods, our QLBM more accurately models Newtonian viscous flow, avoids post-selection, and reduces the measurement cost relative to Carleman-based and hybrid methods, though it requires additional qubits for implementing the H-step (see Sec. 6 and Tab. S1 in Supplementary Information).

DATA AVAILABILITY

The datasets generated during the current study are available in Zenodo at <https://doi.org/10.5281/zenodo.17231277>.

CODE AVAILABILITY

Simulation source codes are available in QLBM at <https://github.com/YYgroup/QLBM>.

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AUTHOR CONTRIBUTIONS

Y.Y. and B.W. designed research. B.W. and Z. M. performed research. All authors contributed to the analysis of data and writing of the manuscript.

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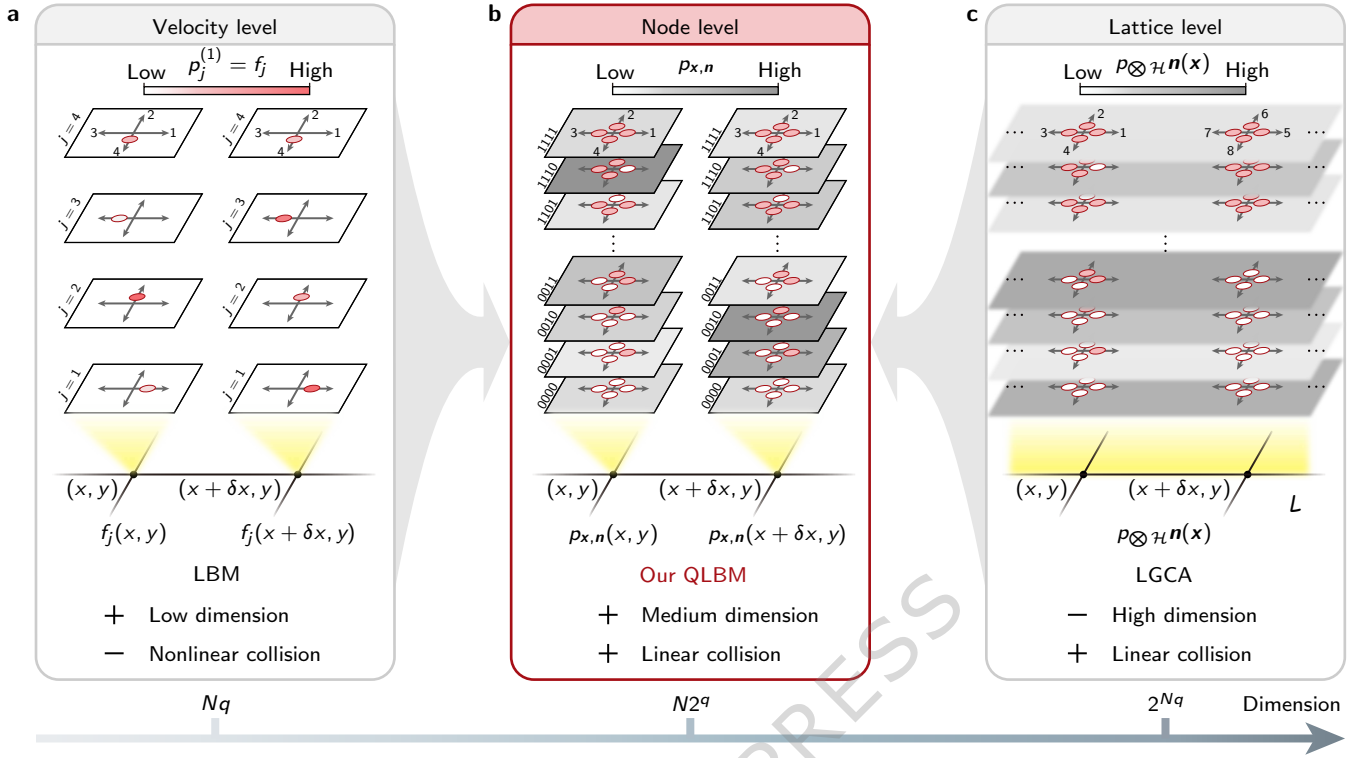


FIG. 1. Schematic of three-level ensemble descriptions for lattice gas: (a) velocity-level ensemble, corresponding classical LBM; (b) node-level ensemble, corresponding to our QLBM; (c) lattice-level ensemble, corresponding to LGCA. The lattice $\mathcal{L}$ comprises N discrete nodes (black dots). Each node represents a local flow state, and it consists of $q = 4$ cells (red circles) for the D2Q4 model. Each card (containing nodes) represents a realization. The probability of each realization is depicted by the saturation level of circles in (a) and the saturation level of cards in (b) and (c). Our QLBM is featured by the medium dimensionality (computational cost) and linear collision treatment (quantum compatibility), suitable for an efficient quantum algorithm. The particle collision in LBM in (a) involves nonlinear calculations with probability values for different cells, whereas the collision in our QLBM in (b) or LGCA in (c) is characterized by a linear combination of different realizations.

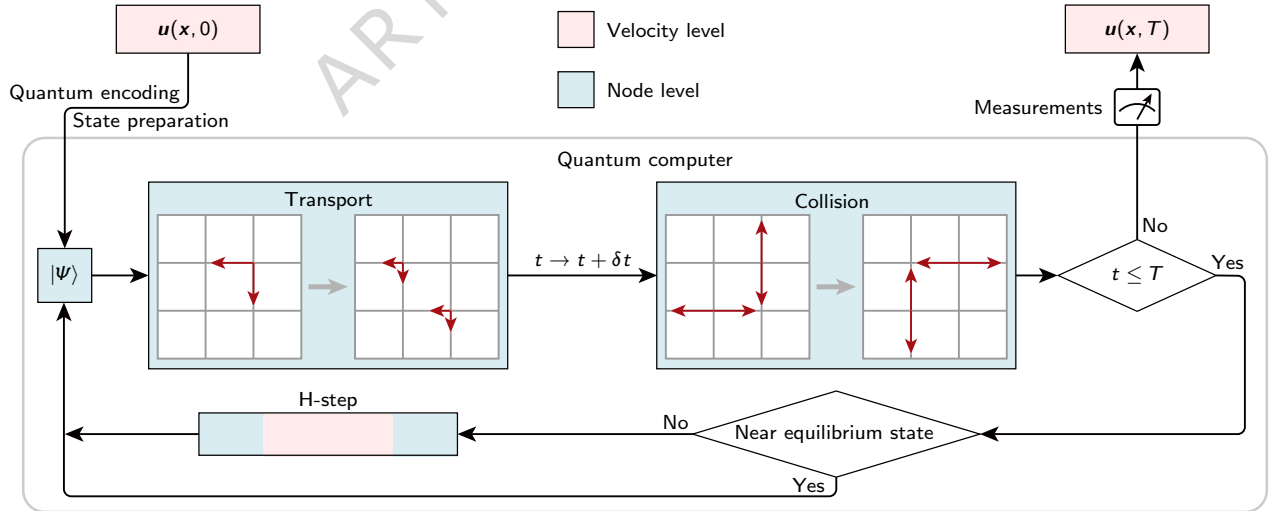


FIG. 2. Flowchart for our QLBM. The velocity field u is embed into the amplitudes of the quantum state $|\psi\rangle$ via probability encoding. The preparation, measurement, and time evolution of the quantum state's density matrix is preformed on a quantum computer. When the state deviates significantly from equilibrium (quantified by the H-function), the H-step relaxes the ensemble back to equilibrium. During the H-step, the state density matrix alternates between velocity-level and node-level ensembles. The state is measured to reconstruct the velocity field when the simulation is ended at a given time $t = T$.

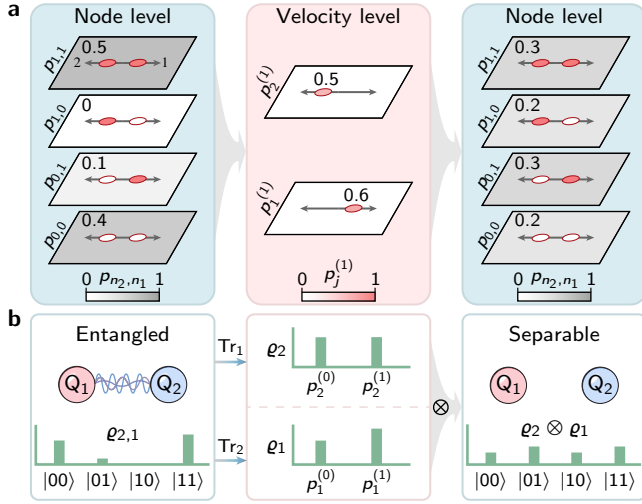


FIG. 3. (a) Schematic for inter-ensemble transformations in the H-step relaxing the ensemble toward equilibrium. The color code and symbol meaning are the same as in Fig. 1. A node-level ensemble (left panel), consisting of two cells and four realizations, can be transformed into a velocity-level ensemble (middle panel) with two realizations under mass conservation, where $p_1^{(1)} = p_{0,1} + p_{1,1}$ and $p_2^{(1)} = p_{1,0} + p_{1,1}$ are calculated by Eq. (4). Assuming independence among cells, the velocity-level ensemble can be transformed back to node-level one (right panel), where $p_{0,0} = p_1^{(0)} p_2^{(0)}$, $p_{0,1} = p_1^{(1)} p_2^{(0)}$, $p_{1,0} = p_1^{(0)} p_2^{(1)}$, and $p_{1,1} = p_1^{(1)} p_2^{(1)}$ are calculated by Eq. (5). Through these two transformations, the effective dimensionality is reduced from four to two, leading to smoothing of the probability distribution (right panel). (b) Schematic for the quantum representation of the H-step. The histogram plots the probability distribution. The two entangled qubits Q_1 and Q_2 (left panel), with density matrix $\varrho_{2,1}$ encoding p_{n_2, n_1} , can be reduced to a single qubit (middle panel) via the partial trace Tr_1 or Tr_2 . Performing this operation on two identical $\varrho_{2,1}$ yields two independent qubits, with density matrices ϱ_1 and ϱ_2 . These two qubits form a separable state $\varrho_2 \otimes \varrho_1$ (right panel). In the H-step, the entangled qubits are disentangled, while the measurement outcomes of each qubit remain unchanged.

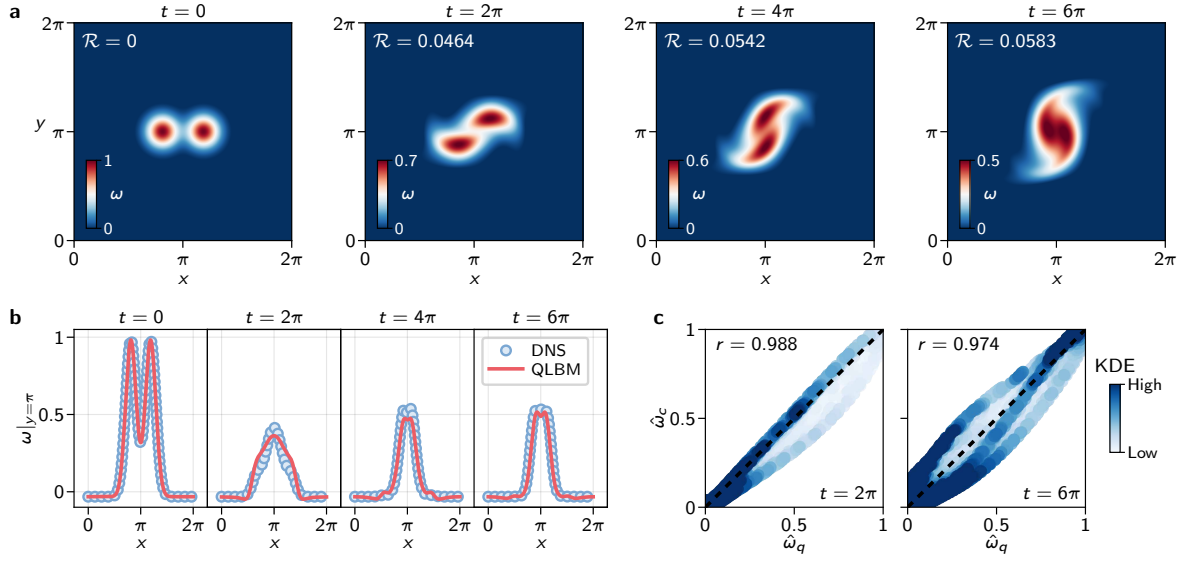


FIG. 4. Comparison between the results of QLBM and DNS for the 2D vortex-pair merging at $t = 0, 2\pi, 4\pi$, and 6π with $\text{Re} = 350$ on 2048^2 nodes. (a) Vorticity contours illustrate the nonlinear interaction of two merging vortices, with the normalized L^2 -norm error $\mathcal{R}$. (b) Cross-sectional vorticity profiles along $y = \pi$ from QLBM and DNS results. (c) Scatter plots of normalized vorticities $\hat{\omega}_q$ for QLBM and $\hat{\omega}_c$ for DNS, with correlation coefficients shown in the upper left. Data points are color-coded by the KDE.

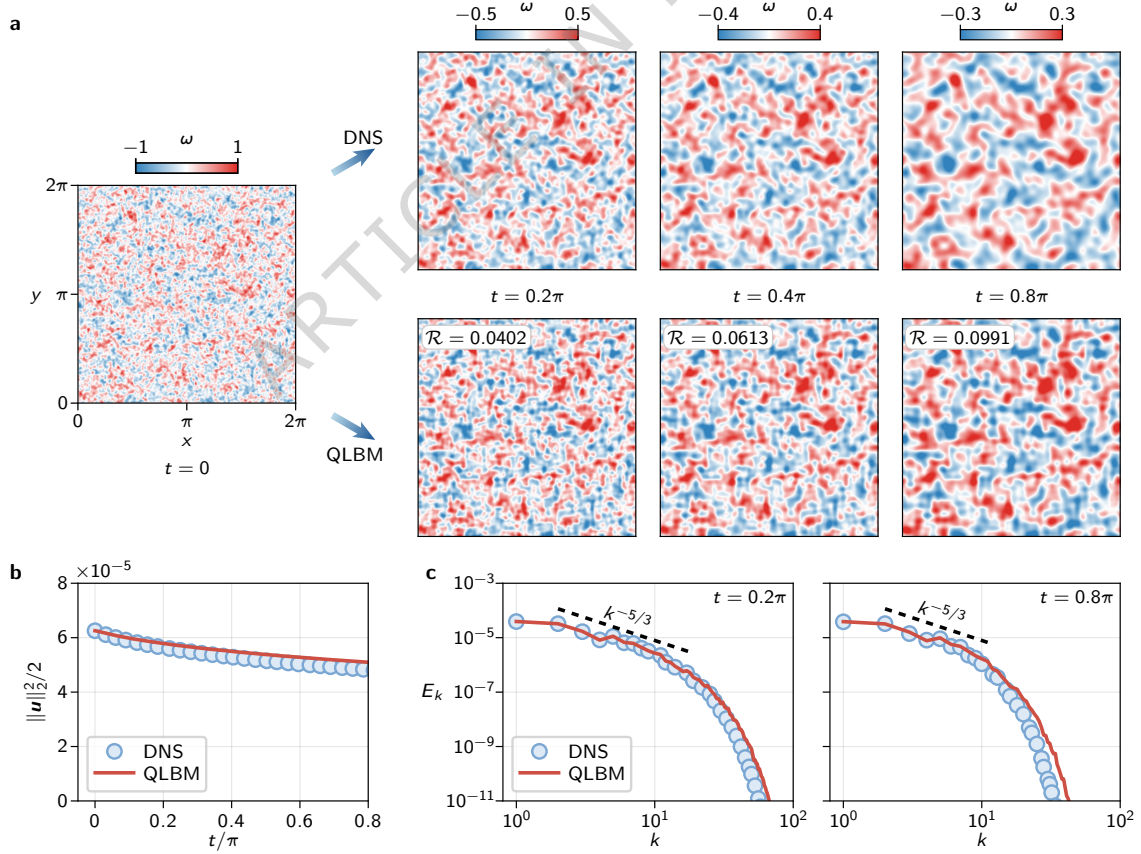


FIG. 5. Comparison between the results of QLBM and DNS for 2D decaying homogeneous isotropic turbulence with $\text{Re} = 102$ on 4096^2 nodes. (a) Vorticity contours at $t = 0.2\pi, 0.4\pi$, and 0.8π . The normalized L^2 -norm error $\mathcal{R}$ is labeled for the QLBM results. (b) Evolution of the average energy. (c) Energy spectra at $t = 0.2\pi$ and 0.8π .